Additive Clocks, Non-Reversible Recurrences, and Transcendental Prime-Representing Constants

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Abstract

For integers $a \geq 2$, $b \geq 1$ and $\varepsilon \in \{-1, 1\}$, the sequence $ba^n + \varepsilon n$ visits every residue class modulo every positive integer with positive natural density. We prove this by induction on the modulus, lowering an eventual period to its greatest common divisor with the modulus. The sequences admit an inhomogeneous recurrence of order two but no reversible integer recurrence, even after changing the order. Pulling back a theorem of Dubickas gives positive lower density of simultaneous composite values in each fixed integer window about $\lfloor \alpha^{ba^n+\varepsilon n} \rfloor$ for every Pisot or Salem number α. This supplies sequences requested in Saito's Problem 1.7. A separate application of Saito's Type-B criterion proves transcendence of the least prime-representing constants for $ba^k \pm k$ when $a \geq 3$, and for two carefully indexed alternating-sign sequences. For the negative-sign family, we also prove an exponentially sharp asymptotic about the algebraic anchor $2^{1/(a-1)}$. The anchor itself fails infinitely many prime conditions. The manuscript distinguishes proved applications from the external transcendence criterion and makes no claim to resolve Mills' constant or to replace prime search by a numerical approximation.

1 The questions and the organizing principle

The non-reversible recurrence problem asks for an integer sequence $R(n)$, governed by a linear recurrence with a constant inhomogeneous term, such that

$$\left\lfloor \alpha^{R(n)} \right\rfloor$$

is composite infinitely often for every Pisot number α. This is Problem 1.7 in Saito's preprint [1]. The sequence is to be chosen independently of α. Non-reversibility means that the trailing recurrence coefficient cannot be 1 or -1; below we prove the stronger fact that no other recurrence representation makes our sequences reversible.

Our examples are

$$R_{a,b,\varepsilon}(n) = ba^n + \varepsilon n, \qquad a \geq 2,\ b \geq 1,\ \varepsilon \in \{-1, 1\}. \tag{1}$$

The same elementary congruence property has another application. For a positive integer sequence $C = (C_k)_{k\geq 1}$, define

$$\mathcal{W}(C) = \{x > 1 : \left\lfloor x^{C_k} \right\rfloor \in \mathcal{P} \text{ for all } k \geq 1\}, \qquad \xi(C) = \min \mathcal{W}(C) \tag{2}$$

whenever the minimum exists; $\mathcal{P}$ is the set of positive primes. For the exponent sequences $C_k = ba^k + \varepsilon k$, $a \geq 3$, the congruence property verifies all arithmetic assumptions of Saito's Type-B theorem [2]. It follows that every one of these specified minima exists and is transcendental.

The term *additive clock* refers simply to the linear term εn. When the exponential part returns modulo q, this term moves by εt, where t is an eventual period. Although this increment may fail to generate all residues modulo q, it generates all residues compatible modulo $\gcd(t, q)$. The latter modulus is strictly smaller. Induction completes the missing residues.

The general congruence theorem is the shared input, not a new proof of Saito's transcendence theorem. We give proofs of the congruence, recurrence, pullback, braid and asymptotic statements, while explicitly citing the two external number-theoretic inputs. Source identifiers T47–T49 and T56–T57 refer to the author's initial research notes. The special case T48 is absorbed into the stronger coefficient theorem.

2 Every residue occurs with positive density

For $S \subseteq \mathbb{N}$, let $\mathrm{d}(S)$ denote its natural density when that density exists, and let $\underline{\mathrm{d}}(S)$ denote its lower natural density.

Theorem 2.1 (Universal residues; [source T47], [source T56(a)])**.** *Fix the parameters in* (1)*. For every integer $q \geq 1$ and every $c \in \mathbb{Z}$, the set*

$$S(q, c) = \{n \geq 1 : R_{a,b,\varepsilon}(n) \equiv c \pmod{q}\}$$

has positive natural density. It is an eventually periodic subset of $\mathbb{N}$, and it contains all sufficiently large members of at least one arithmetic progression.

Lemma 2.2 (A short eventual period)**.** *Let $a \geq 2$ and $q > 1$. There exist positive integers e, t such that*

$$a^{n+t} \equiv a^n \pmod{q} \quad (n \geq e), \qquad 1 \leq t < q. \tag{3}$$

Proof. Write $q = uv$, where every prime divisor of u divides a and $\gcd(a, v) = 1$. More precisely, u is the product of the prime-power factors of q supported on primes dividing a. Thus $\gcd(u, v) = 1$. For all sufficiently large n, $u \mid a^n$. If $v > 1$, choose $t = \mathrm{ord}_v(a)$. Then

$$t \leq \varphi(v) < v \leq q,$$

and (3) follows from the Chinese remainder theorem. If $v = 1$, choose $t = 1 < q$, since the exponential part is eventually zero modulo q. □

Proof of Theorem 2.1. We first prove the existence of arbitrarily large solutions, using strong induction on q. For $q = 1$ every index works. Suppose $q > 1$, and take e, t as in Lemma 2.2. Set

$$g = \gcd(t, q) < q.$$

By the induction hypothesis, there are arbitrarily large $n \geq e$ with $R_{a,b,\varepsilon}(n) \equiv c \pmod{g}$. For such an n, the linear congruence

$$\varepsilon j t \equiv c - R_{a,b,\varepsilon}(n) \pmod{q} \tag{4}$$

is solvable because its right-hand side is divisible by g. Choose a nonnegative solution j and put $N = n + jt$. The period identity gives $a^N \equiv a^n \pmod{q}$, so

$$R_{a,b,\varepsilon}(N) \equiv R_{a,b,\varepsilon}(n) + \varepsilon j t \equiv c \pmod{q}.$$

Since $N \geq n$, solutions can be chosen arbitrarily large. The coefficient b is unchanged throughout and imposes no coprimality condition.

Now put $Q = \operatorname{lcm}(q, t)$. For every $n \geq e$,

$$R_{a,b,\varepsilon}(n+Q) \equiv R_{a,b,\varepsilon}(n) \pmod{q}.$$

Thus $S(q, c)$, apart from finitely many indices, is a union of residue classes modulo Q. At least one such class is present by the first part of the proof. Consequently its density exists and is at least $1/Q > 0$. □

Remark 2.3 (Surjectivity is not uniform distribution). The theorem supplies at least one eventual index class for each target residue. It does not assert that the q residues have equal density. The period Q may depend on a and q, and different residues can have different numbers of representatives modulo Q.

3 Intrinsic non-reversibility

An integer sequence R is an inhomogeneous linear recurrence sequence of order d if, for fixed integers $a_0, \ldots, a_{d-1}, c$ with $a_0 \neq 0$,

$$R(n+d) = a_{d-1}R(n+d-1) + \cdots + a_0 R(n) + c.$$

Such a representation is reversible when $a_0 = \pm 1$. In a sequence with several possible recurrence representations, merely displaying one with $a_0 \neq \pm 1$ does not exclude a hidden reversible one. We therefore check all integer representations at once.

Proposition 3.1. *The sequence in* (1) *is positive and strictly increasing for* $n \geq 1$. *It satisfies the order-two relation*

$$R(n+2) = (a+1)R(n+1) - aR(n) + \varepsilon(1-a). \tag{5}$$

Every integer inhomogeneous recurrence representation of this sequence has trailing coefficient divisible by a. *In particular, none is reversible.*

Proof. The exponential and linear terms directly verify (5). Positivity follows from $ba^n - n \geq 2^n - n \geq 1$. Also

$$R(n+1) - R(n) = b(a-1)a^n + \varepsilon \geq 1.$$

Let E be the shift operator, and suppose $Q(E)R$ is constant for some monic integer polynomial Q associated to an inhomogeneous recurrence. The exponential term in $Q(E)R$ is $ba^n Q(a)$; the coefficient of n is $\varepsilon Q(1)$. An exponential plus a polynomial can be constant for all positive n only if

$$Q(a) = 0, \qquad Q(1) = 0.$$

Since $a \neq 1$, the monic polynomial $(X-a)(X-1)$ divides Q in $\mathbb{Z}[X]$. Thus $a \mid Q(0)$. The trailing coefficient of the recurrence is $-Q(0)$ and therefore cannot be 1 or -1. This also shows directly that the homogeneous minimal annihilator is $(X-a)(X-1)^2$. □

The term "order two" in Proposition 3.1 refers to a recurrence with a constant inhomogeneous term. Its homogeneous annihilator has degree three. Conflating these two conventions would misstate the non-reversibility conclusion.

4 Pulling back composite progressions

We use the following established result as an external input. It is Theorem 1.1 of Dubickas [3], also recorded as (D1) in [1].

Theorem 4.1 (Dubickas, external input)**.** *Let $(T_m)_{m\ge1}$ be an unbounded integer linear recurrence sequence. For each integer $J \ge 1$, there are positive integers L, m_0 and primes p_j, $|j| \le J$, such that*

$$p_j \mid T_{Lk+m_0} + j \qquad (k \ge 1,\ |j| \le J).$$

The primes p_j need not be distinct.

We pass from trace sequences to floors with a bounded-error argument. This step only requires a finite collection of available divisors. It does not require one particular prime to divide every floor value with the same external shift.

Lemma 4.2 (Composite floor windows)**.** *Let α be a Pisot or Salem number, and fix $H \ge 0$. There exist positive integers L, m_0, K such that*

$$\left\lfloor \alpha^{Lk+m_0} \right\rfloor + h \quad \text{is composite for all } k \ge K, \quad |h| \le H. \tag{6}$$

Proof. Let d be the degree of α and let $\alpha = \alpha_1, \dots, \alpha_d$ be its conjugates. The integer trace sequence

$$T_m = \sum_{i=1}^{d} \alpha_i^m$$

satisfies the linear recurrence given by the monic minimal polynomial. For a Pisot number the other conjugates have modulus less than 1; for a Salem number they have modulus at most 1. Hence

$$|T_m - \alpha^m| \le d - 1, \qquad e_m := \lfloor \alpha^m \rfloor - T_m \in \mathbb{Z}, \quad |e_m| \le d.$$

The same bound shows $T_m \to \infty$, so the trace recurrence is unbounded. Apply Theorem 4.1 with $J = H + d$. Along the resulting progression, every $T_{Lk+m_0} + j$, $|j| \le J$, has the fixed divisor p_j. For sufficiently large k, all these positive integers exceed all the finitely many primes p_j. Finally,

$$\left\lfloor \alpha^{Lk+m_0} \right\rfloor + h = T_{Lk+m_0} + (e_{Lk+m_0} + h),$$

and $|e_{Lk+m_0} + h| \le J$. Choosing the available prime for this integer shift proves proper divisibility and (6). □

Theorem 4.3 (A solution of the non-reversible construction problem; [source T49])**.** *For every $a \ge 2$, $b \ge 1$, $\varepsilon = \pm1$, every Pisot or Salem number α, and every integer $H \ge 0$, the set*

$$\left\{ n \ge 1 : \left\lfloor \alpha^{ba^n + \varepsilon n} \right\rfloor + h \text{ is composite for all } h \in \mathbb{Z},\ |h| \le H \right\}$$

has positive lower natural density.

Proof. Choose L, m_0, K from Lemma 4.2. By Theorem 2.1, the set

$$S = \{ n \ge 1 : R_{a,b,\varepsilon}(n) \equiv m_0 \pmod{L} \}$$

has positive natural density. Since $R_{a,b,\varepsilon}(n) \to \infty$, all sufficiently large $n \in S$ satisfy $R_{a,b,\varepsilon}(n) = Lk_n + m_0$ with $k_n \ge K$. Equation (6) applies to each of them. Removing finitely many indices from S leaves its density unchanged, proving the claimed lower density. □

Taking $b = 1$, $H = 0$ and restricting to Pisot numbers gives the exact construction asked for in Saito's Problem 1.7. The clock is independent of α, and Proposition 3.1 verifies the required non-reversibility. The Salem and fixed-window statements are further applications of the same proof. The theorem does not replace the exponent 3^r in the separate cubic Pisot problem by an equivalent one: $a^r \pm r$ is a different sequence.

5 The external Type-B criterion, with all hypotheses

The following is the transcendence conclusion of Theorem 2.3 in Saito's version 3 [2], expressed directly in terms of the exponent sequence. This is an external theorem, not a lemma proved in this manuscript.

Criterion 5.1 (Saito Type B). *Let $C_k \in \mathbb{N}$, set $c_1 = C_1$ and $c_{k+1} = C_{k+1}/C_k$, and assume*

$$\begin{aligned} &\text{(B1)} \quad c_1 \geq 1, \qquad \text{(B2)} \quad c_{k+1} \geq 2 \ (k \geq 1), \\ &\text{(B3)} \quad \limsup_{k\to\infty} c_{k+1} > 40/19, \qquad \text{(B4)} \quad C_k = c_1 \cdots c_k \in \mathbb{N}. \end{aligned}$$

Choose sufficiently small $\eta > 0$ so that $I = \{k : c_{k+1} \geq 40/19 + \eta\}$ is infinite. Assume also

$$\begin{aligned} &\text{(B5)} \quad \forall m \geq 1 \ \exists k \in I, \quad k > m, \quad C_m \mid C_k; \\ &\text{(B6)} \quad \forall m, L \geq 1 \ \exists k \in I, \quad k > m, \quad C_k \equiv C_m \pmod{LC_m}. \end{aligned}$$

Then $\xi(C)$ exists and is transcendental.

The existence of a minimum causes no extra difficulty once the prime-representing set is nonempty. We include the basic order argument, which is also useful for understanding the exact endpoints of the intervals used later.

Lemma 5.2 (Existence of a minimum from nonemptiness). *If C_k are positive integers and $\mathcal{W}(C) \neq \varnothing$, then $\mathcal{W}(C)$ has a minimum.*

Proof. Let $x = \inf \mathcal{W}(C)$. The first prime condition gives $x \geq 2^{1/C_1} > 1$. There is a sequence $x_j \in \mathcal{W}(C)$ with $x_j \geq x$ and $x_j \to x$. For each fixed k, the values $x_j^{C_k}$ approach x^{C_k} from above. For all sufficiently large j, their floors equal $\left\lfloor x^{C_k} \right\rfloor$, including when x^{C_k} itself is an integer. This eventually constant floor is prime because every $x_j \in \mathcal{W}(C)$. Thus $x \in \mathcal{W}(C)$. □

Criterion 5.1 is used only after checking every one of its hypotheses. In particular, a congruence result at arbitrarily large indices does not automatically place those indices in I. Here we will have I cofinite, making the required intersection explicit.

6 Transcendence for all coefficients and both signs

Theorem 6.1 (Coefficient family; [source T48], [source T56(a)]). *For every integer $a \geq 3$, every integer $b \geq 1$, and both choices of $\varepsilon \in \{-1, 1\}$, the constant*

$$\xi\big((ba^k + \varepsilon k)_{k\geq 1}\big)$$

exists and is transcendental.

Proof. Put $C_k = ba^k + \varepsilon k$. Then $C_1 = ba + \varepsilon \geq 2$, and every C_k is a positive integer. Moreover,

$$C_{k+1} - 2C_k = b(a-2)a^k + \varepsilon(1-k) > 0. \tag{7}$$

For $\varepsilon = -1$ both terms on the right are nonnegative and the first is positive. For $\varepsilon = 1$, use $b(a-2)a^k \geq 3^k > k-1$. This verifies (B1), (B2) and (B4). Also

$$\frac{C_{k+1}}{C_k} \longrightarrow a > 40/19,$$

so (B3) holds. Fix any sufficiently small $0 < \eta < a - 40/19$. Then I in Criterion 5.1 is cofinite.

For a given m, apply Theorem 2.1 with modulus C_m and target 0. It supplies arbitrarily large k with $C_m \mid C_k$. Choose one with $k > m$ and $k \in I$, verifying (B5). For given m, L, apply the same theorem with modulus LC_m and target C_m. Again choose a sufficiently large solution in I; this gives (B6). Criterion 5.1 proves the assertion. □

For $a = 3$, $b = 1$, the first few exponents are

$$(3^k - k)_{k\geq 1} = (2, 7, 24, 77, 238, \ldots),$$

$$(3^k + k)_{k\geq 1} = (4, 11, 30, 85, 248, \ldots).$$

These are two different exponent sequences defining two minima. Their transcendence follows from the same theorem, but this does not establish that the minima are unequal or order them relative to Mills' constant.

Corollary 6.2 (Adjacent coefficient lines)**.** *Let $A_k = 3^k - k$ and $B_k = 3^k + k$. For every integer $r \geq 0$, both exponent sequences*

$$(r+1)A_k + rB_k = (2r+1)3^k - k,$$

$$rA_k + (r+1)B_k = (2r+1)3^k + k$$

define transcendental least prime-representing constants.

Proof. Apply Theorem 6.1 with $a = 3$, $b = 2r + 1$ and the corresponding sign. □

The coefficient theorem also covers even b, which are not obtained from these adjacent nonnegative coefficient pairs. Nothing in the congruence proof depended on b being odd.

7 Alternating signs and the necessary indexing

Define

$$P_k = 3^k + (-1)^k k, \qquad Q_k = 3^k - (-1)^k k. \tag{8}$$

The notation P_k here is an exponent sequence, not a prime. We spell out the starting indices because one of the sequences fails a Type-B growth condition at its first step.

Lemma 7.1 (Residues of the braids)**.** *Each sequence in (8) visits every residue class modulo every positive integer with positive natural density.*

Proof. For either fixed sign ε, one has $3^k + \varepsilon k \equiv 1 + k \pmod 2$. First suppose q is even and a residue c modulo q is given. The required index parity is $t \equiv c - 1 \pmod 2$. On that parity, $P_k = 3^k + \varepsilon k$ with $\varepsilon = (-1)^t$. Theorem 2.1 for this fixed sign supplies positive-density solutions modulo q, and the congruence modulo 2 forces every one of them to have parity t. They are therefore solutions for P_k. For Q_k, choose $\varepsilon = -(-1)^t$ instead.

If q is odd, choose a lift c' modulo $2q$ with $c' \equiv c \pmod q$ and any prescribed parity. Apply the even-modulus argument to $2q$. This supplies a positive-density subset of the desired solutions modulo q. Finally, both braid sequences are eventually periodic modulo q: combine an eventual period of 3^k with the periods q of k and 2 of $(-1)^k$. Thus their full solution sets have natural densities, which must be positive. □

Theorem 7.2 (Braid constants; [source T56(b)])**.** *Both*

$$\xi\big((P_k)_{k\ge 1}\big) \quad \textit{and} \quad \xi\big((Q_k)_{k\ge 2}\big)$$

exist and are transcendental. In the second expression the sequence is reindexed from 1*, namely* $C_j = Q_{j+1}$.

Proof. Direct subtraction gives

$$P_{k+1} - 2P_k = 3^k - (-1)^k(3k+1),$$
$$Q_{k+1} - 2Q_k = 3^k + (-1)^k(3k+1).$$

For $k \ge 2$, the inequality $3^k > 3k + 1$ follows by induction, so both differences are positive. Also $P_1 = 2$, $P_2 = 11$ satisfy the first growth condition for the P sequence. For the Q sequence, $Q_1 = 4$, $Q_2 = 7$ give $Q_2 < 2Q_1$, explaining the stated truncation. The reindexed sequence starts at $Q_2 = 7$ and has all subsequent ratios greater than 2. Both ratios tend to 3.

Consequently (B1)–(B4) hold for the two stated sequences, and I is cofinite. Lemma 7.1, unaffected by deleting finitely many indices, verifies (B5) and (B6) exactly as in the proof of Theorem 6.1. Apply Criterion 5.1. □

The proof makes no assertion about the minimum defined by the full Q sequence starting at $k = 1$. Removing a prime condition can change the minimum, so a tail theorem must not be silently promoted to the full sequence.

8 Scaling, synchronization and elementary combinations

Proposition 8.1 (Exact scaling; [source T56(c)])**.** *If C is a positive integer sequence with $\mathcal{W}(C) \neq \varnothing$ and $d \ge 1$ is an integer, then*

$$\xi\big((dC_k)_{k\ge 1}\big) = \xi(C)^{1/d}.$$

Proof. For $x > 1$,

$$x \in \mathcal{W}(dC) \quad \Longleftrightarrow \quad x^d \in \mathcal{W}(C).$$

The map $x \mapsto x^d$ is an increasing bijection of $(1, \infty)$. It therefore carries the minimum of the first set to the minimum of the second, which exists by Lemma 5.2. □

For $A_k = 3^k - k$, $B_k = 3^k + k$ and $M = \xi((3^k)_{k\ge 1})$, Proposition 8.1 gives

$$\xi\big((A_k + B_k)_{k\ge 1}\big) = \sqrt{M}. \tag{9}$$

This is an identity about the *sum of exponent sequences*. It is not an identity involving $\xi(A)+\xi(B)$ or $\xi(A)\xi(B)$. The transcendence of the left side is equivalent to that of M; the separate transcendence statements for $\xi(A)$ and $\xi(B)$ do not decide it.

Proposition 8.2 (A simultaneous-return obstruction; [source T56(d)]). *For every $m \geq 1$, no integer $k > m$ satisfies both*

$$A_k \equiv A_m \pmod{3^{m+1}}, \qquad B_k \equiv B_m \pmod{3^{m+1}}.$$

Proof. Since 2 is invertible modulo 3^{m+1}, adding the two congruences would give $3^k \equiv 3^m \pmod{3^{m+1}}$. But $k \geq m+1$ makes the left side zero and the right side nonzero modulo this modulus. □

The proposition does not contradict individual Type-B returns. Their indices need not coincide. It rules out this particular attempt to synchronize the two returns and cancel the linear terms, rather than all possible relations between the two constants.

Corollary 8.3 (Elementary combinations; [source T56(e)]). *Let $\alpha = \xi(A)$ and $\beta = \xi(B)$. At least two of*

$$\alpha+\beta, \qquad \alpha\beta, \qquad \alpha/\beta$$

are transcendental. Also $\alpha + i\beta$ is transcendental.

Proof. Both α and β are positive transcendental numbers by Theorem 6.1. If their sum and product were algebraic, they would be roots of a quadratic polynomial with algebraic coefficients. If their sum and ratio were algebraic, then $\beta = (\alpha+\beta)/(1+\alpha/\beta)$ would be algebraic. If their product and ratio were algebraic, then $\alpha^2 = (\alpha\beta)(\alpha/\beta)$ would be algebraic. Every case contradicts the transcendence of α or β. Finally, the real and imaginary parts of an algebraic complex number are algebraic, excluding algebraicity of $\alpha + i\beta$. □

These are algebraic consequences of the two known transcendental values. They do not imply algebraic independence.

9 An algebraic anchor that fails infinitely many conditions

For the rest of the manuscript let

$$C_k = a^k - k, \quad \xi_a = \xi(C), \quad R_a = 2^{1/(a-1)}, \quad r_a = \frac{a^2-2}{a-1} = a+1-\frac{1}{a-1}, \qquad a \geq 3. \tag{10}$$

The constant R_a is the smallest number compatible with a first prime value at the first exponent. It is a useful scale, but it does not satisfy the full prime conditions.

Proposition 9.1 (Exact anchor obstruction; [source T57(a)]). *For every integer $a \geq 3$,*

$$\xi_a > R_a.$$

For every $k > 1$ with $k \equiv 1 \pmod{a-1}$, the value $R_a^{C_k}$ is an integer power of 2 with exponent greater than 1, and hence its floor is composite.

Proof. The first prime condition gives $\xi_a^{a-1} \geq 2$, so $\xi_a \geq R_a$. Since $a \equiv 1 \pmod{a-1}$,

$$C_k \equiv 1-k \pmod{a-1}.$$

At the stated indices $m_k = C_k/(a-1)$ is a positive integer greater than 1. Therefore $R_a^{C_k} = 2^{m_k}$ is composite. Such indices are infinite, so $R_a \notin \mathcal{W}(C)$ and the lower bound for its minimum is strict. □

This strict inequality is independent of the transcendence proof. For example, $R_3 = \sqrt{2}$ gives floors 2 and 11 at exponents 2 and 7, but at the next exponent 24 it gives 4096. Agreement with the first two prime conditions therefore provides no all-index certificate.

10 Short prime intervals and nested prime cylinders

The analytic external input is the Baker–Harman–Pintz prime interval theorem [4], in the right-interval form also stated in [2, Theorem 1.6]. With

$$\theta = 21/40,$$

there is a constant X_0 such that, for every $X \geq X_0$, a prime exists strictly between X and $X + X^\theta$, with the upper endpoint allowed. The usual quantitative version gives many primes, so excluding a possible prime at X does not alter this assertion for sufficiently large X. No explicit numerical X_0 is used here.

Lemma 10.1 (Room for the next cylinder)**.** *For every $p \geq 2$ and every real $r \geq 3$,*

$$(p+1)^r - p^r > p^{\theta r} + 1.$$

Proof. The integral form of the mean value estimate gives $(p+1)^r - p^r > rp^{r-1}$. Since

$$r - 1 - \theta r = (19/40)r - 1 \geq 17/40 > 0,$$

one has $rp^{r-1} \geq 3p^{\theta r} > p^{\theta r} + 1$. □

Lemma 10.2 (A branch beginning at 2)**.** *There exists an integer a_0 such that, for every integer $a \geq a_0$, a number $A_a \in \mathcal{W}((a^k - k)_{k\geq 1})$ can be chosen with*

$$\left\lfloor A_a^{a-1} \right\rfloor = 2.$$

Writing $X = 2^{r_a}$, it can be chosen so that

$$R_a < A_a < (X + X^\theta + 1)^{1/(a^2-2)}. \tag{11}$$

Proof. Put $r_k = C_{k+1}/C_k$. Direct calculation gives

$$r_k = a + \frac{(a-1)k - 1}{a^k - k} > a \geq 3. \tag{12}$$

Choose a_0 so that $2^{r_a} \geq X_0$ for $a \geq a_0$; this is possible because $r_a \to \infty$. Start with $p_1 = 2$ and choose recursively a prime satisfying

$$p_k^{r_k} < p_{k+1} \leq p_k^{r_k} + (p_k^{r_k})^\theta.$$

The inputs to the prime interval theorem remain above X_0: p_k strictly increases and $r_k \geq 3$, while the initial input is 2^{r_a}; more directly, for $k \geq 2$, $p_k > 2^{r_a}$. Lemma 10.1 gives

$$p_{k+1} + 1 < (p_k + 1)^{r_k}.$$

Consequently the compact intervals

$$I_k = [p_k^{1/C_k}, (p_k+1)^{1/C_k}]$$

satisfy $I_{k+1} \subset \mathrm{int}(I_k)$. Their nonempty nested intersection contains a point A_a. For each fixed k, this point belongs to I_{k+1} and therefore to the interior of I_k. It follows that $\left\lfloor A_a^{C_k} \right\rfloor = p_k$ for every k. In particular its first floor is 2.

At $k = 1$, the upper bound on the chosen second prime is $p_2 \leq X + X^\theta$. Since A_a lies strictly inside I_2, $A_a < (p_2 + 1)^{1/(a^2-2)}$, proving the upper bound in (11). The lower bound follows either from strict containment in I_1 or from Proposition 9.1. □

11 The transcendental boundary layer

Theorem 11.1 (Root normalization; [source T57(b)–(c)]). *For all sufficiently large integers a,*

$$0 < \log \frac{\xi_a}{R_a} \le \frac{\log(1 + 2^{-19r_a/40} + 2^{-r_a})}{a^2 - 2}, \qquad \left\lfloor \xi_a^{a-1} \right\rfloor = 2. \tag{13}$$

As $a \to \infty$ through integers,

$$\xi_a = 2^{1/(a-1)} \left(1 + O\left(\frac{2^{-19a/40}}{a^2}\right)\right), \tag{14}$$

$$\Theta_a := \xi_a^{a-1} = 2 \left(1 + O\left(\frac{2^{-19a/40}}{a}\right)\right), \tag{15}$$

$$(a-1)(\xi_a - 1) \longrightarrow \log 2. \tag{16}$$

For every integer $a \ge 3$, Θ_a is transcendental and $\Theta_a > 2$.

Proof. By minimality and Lemma 10.2, $\xi_a \le A_a$ for all $a \ge a_0$. The branch has $A_a < 3^{1/(a-1)}$, while Proposition 9.1 gives $\xi_a > 2^{1/(a-1)}$. Thus the minimum also has first floor 2.

Let $X = 2^{r_a}$. Since $X^{1/(a^2-2)} = R_a$, (11) implies

$$1 < \frac{\xi_a}{R_a} < (1 + X^{-19/40} + X^{-1})^{1/(a^2-2)}.$$

Taking logarithms proves (13) (and a strict upper bound would also be valid). Because $r_a = a + 1 - 1/(a-1)$ and $\log(1+u) \le u$ for $u \ge 0$,

$$\log(\xi_a / R_a) = O(2^{-19a/40}/a^2).$$

Exponentiating gives (14), and multiplying this logarithmic estimate by $a - 1$ gives (15). Finally, $2^{1/(a-1)} = 1 + \log 2/(a-1) + O(a^{-2})$, so (16) follows.

Theorem 6.1 proves ξ_a transcendental for every $a \ge 3$, not merely for large a. A positive integer power of a transcendental number is transcendental: otherwise it would make the original number algebraic over $\overline{\mathbb{Q}}$. Thus Θ_a is transcendental. Its strict lower bound follows from Proposition 9.1. □

The convergence in (15) supplies an explicit family of transcendental numbers approaching the algebraic value 2. It does not suggest that a limit of transcendental numbers must be transcendental. The threshold a_0 comes from the asymptotic prime interval input and is not numerically specified by this proof. In particular, the first-floor assertion in (13) has not been proved here for every $a \ge 3$.

12 Exact correction coordinates and finite prefixes

Define the positive correction coordinate

$$\tau_a = \log(\xi_a^{a-1}/2). \tag{17}$$

Theorem 11.1 gives $\tau_a = O(2^{-19a/40}/a)$. At a given exponent $C_k = a^k - k$, a proposed prime value imposes an exact interval condition.

Proposition 12.1 (Prime-cylinder coordinates; [source T57(d)]). *Let $x = R_a \exp(\tau/(a-1))$, where $\tau \in \mathbb{R}$. For a positive integer p,*

$$\left\lfloor x^{C_k} \right\rfloor = p$$

holds if and only if

$$\frac{a-1}{C_k}\log p - \log 2 \le \tau < \frac{a-1}{C_k}\log(p+1) - \log 2. \tag{18}$$

Proof. The floor equality is equivalent to $p \le x^{C_k} < p+1$. Substitute the expression for x, take logarithms and multiply by $(a-1)/C_k > 0$. □

For a finite prime list $p_1, \dots, p_m$, simultaneous compatibility means that the corresponding half-open intervals (18) have a nonempty intersection. There is no need to rely on rounded root values when certifying their endpoints: for positive integers u, v, d, e,

$$u^{1/d} < v^{1/e} \quad \Longleftrightarrow \quad u^e < v^d.$$

Thus the required comparisons can be reduced to exact integers. The interval width at exponent C_k equals

$$\frac{a-1}{C_k}\log\left(1+\frac{1}{p}\right) \sim \frac{a-1}{C_k p}.$$

It becomes small as the output prime and the exponent grow. A tiny relative correction can therefore change a floor by an integer, and many more output digits can require many more digits of the constant.

For example, when $a = 4$ the first exponents are $3, 14, 61$. The anchor $2^{1/3}$ gives

$$\left\lfloor (2^{1/3})^{14} \right\rfloor = 25 = 5^2.$$

Choosing the prime 29 and putting $x = 29^{1/14}$ gives exactly

$$\left\lfloor x^3 \right\rfloor = 2, \qquad \left\lfloor x^{14} \right\rfloor = 29,$$

since $2^{14} < 29^3 < 3^{14}$. This verifies two prime conditions. It does not identify the global minimum ξ_4 or establish that this particular finite prefix is indefinitely extendible.

13 What the two-stage construction does and does not give

A useful unconditional finite-prefix statement follows from the classical Bertrand theorem. We include its proof to separate it from the asymptotic use of short prime intervals above.

Proposition 13.1 (A two-prime prefix for every $a \ge 3$). *Let $X = 2^{r_a}$, and let p be the smallest prime strictly greater than X. Then*

$$y = p^{1/(a^2-2)} \quad \textit{satisfies} \quad \left\lfloor y^{a-1} \right\rfloor = 2, \qquad \left\lfloor y^{a^2-2} \right\rfloor = p.$$

Proof. Let $m = \lceil X \rceil$. Bertrand's theorem supplies a prime q with $m < q < 2m$. Hence $X < p \le q < 2\lceil X \rceil < 2(X+1)$. Since $r_a \ge 7/2$,

$$(3/2)^{r_a} \ge (3/2)^{7/2} > 4 \quad \Longrightarrow \quad 3^{r_a} > 4X > 2(X+1).$$

Thus $2^{r_a} < p < 3^{r_a}$. Taking r_a-th roots gives $2 < y^{a-1} < 3$, while the other power equals the prime p itself. □

The proposition begins with a selected and certified prime p. It records how that prime is encoded in a two-stage constant. It does not produce a prime without finding one, does not give the exact global minimum, and does not make the uncorrected value $\lfloor 2^{r_a} \rfloor$ prime. For sufficiently large a, the short-interval argument can instead choose the smallest prime above X inside the Baker–Harman–Pintz interval and extend that prefix through nested cylinders. The numerical threshold remains unspecified.

14 Scope and source correspondence

Theorems 2.1 and 4.3 retain the source content T47 and T49, with the coefficient extension from T56. Theorem 6.1 combines T48 and T56(a), so the $b = 1$ special case is not counted as an additional independent method. Theorem 7.2 and Propositions 8.1–8.2 contain the remaining structural portions of T56. Sections 9–13 establish T57 with its large-a quantifiers explicit. The separate source item T53, concerning $\xi(2^k)$, is not part of this manuscript.

The general congruence lemma applies already at $a = 2$, but the transcendence application in Theorem 6.1 is stated only for $a \geq 3$. The limit of consecutive ratios must exceed $40/19$ to invoke the external criterion used here. An endpoint argument for $a = 2$ would require additional input.

The set

$$\{x > 1 : \left\lfloor x^{3^k - k} \right\rfloor \text{ and } \left\lfloor x^{3^k + k} \right\rfloor \text{ are prime for every } k \geq 1\}$$

is not proved nonempty here. The two separate nonempty sets need not intersect. Interleaving their exponent sequences gives adjacent ratios approaching 1 within each pair, so the Type-B growth assumptions cannot simply be inherited. Likewise, no monotonicity of $\xi(C)$ under pointwise comparison of different exponent sequences has been proved.

The logical contribution to Problem 1.7 is an explicit family of intrinsically non-reversible recurrences with universal congruence visitation, combined with an established composite-progression input. The transcendence contribution is verification of a separate, established criterion for new explicit exponent families and their stated braid variants. Neither contribution settles the transcendence of the classical Mills constant, and neither is presented as a proof of historical firstness. All historical priority assertions would require an independent literature audit beyond the proof checks recorded here.

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